

Review: Can Failed Trajectories Reveal Where Agents Go Wrong? Agentic Failure Attribution without Step-Level Labels

Submission to ICLR 2027 (SOAP). I reviewed the manuscript and appendix as submitted. I did not search for the paper or its cited 2026 baselines, to preserve anonymity, so baseline descriptions are taken as given.

Summary

The paper addresses failure attribution for LLM multi-agent systems: given a failed trajectory, identify the decisive step. The proposed method, SOAP, has two parts.

- **Spectral base score.** Mean-pooled hidden states of all steps from a reference set of failed, step-unlabeled trajectories are stacked, and an uncentered SVD is computed. Each test step is scored by the inverse of its projection energy onto a contiguous band of the top-20 right singular vectors.
- **Attention-guided rescoring.** Each step's score is increased by γ times an attention-weighted average of the base scores of later steps that attend to it. This is meant to counter "downstream contamination", where late symptoms outscore the root cause.

On Who&When, CORRECT-Error and TraceElephant with Qwen3.5-9B and DeepSeek-R1-Distill-Llama-8B, SOAP beats same-backbone prompt-based and training-based baselines, by large margins on four of five subsets. The paper also reports ablations, top-k results, cross-dataset transfer, synthetic references and cost.

Strengths

- The problem is timely and the setting is practical: task-level failure is cheap to observe, while step-level blame is not. The method is training-free, needs no judge calls, and can use an open-weight model on logs from proprietary agents.
- Downstream contamination is a clearly articulated failure mode of per-step scorers, and Figure 1 conveys it well.
- The headline numbers are large and consistent across two backbones (for example, 47.62 vs. 39.68 on WW-AG and 34.48 vs. 21.84 on WW-HC with Qwen3.5-9B). The top-k and MRR results show the ranking is useful for triage, not only the argmax.
- The appendix is unusually thorough and candid. It reports seed variance and unfavorable cases (GPT-5 ErrorProbe beating SOAP on WW-AG, $\gamma=0$ being selected on DeepSeek WW-HC). The input format, truncation and search grids are specified well enough to reproduce.

Weaknesses

1. The "without step-level labels" claim does not match the protocol. Gold labels on a 20% validation split select the representation layer (10 options on Qwen), the spectral band (210), the attention band (4), γ (7) and the window w (6). That is on the order of 3×10^5 configurations, chosen on roughly 25 labeled trajectories for WW-AG, 12 for WW-HC and 17-18 for TraceElephant.

- Table 3 shows the result is extremely sensitive to this choice: 39.15 for the selected band vs. 12.70 for the top subspace and 12.17 for the full 20 on WW-AG. The labels are therefore doing substantial work.
- SOAP is nevertheless marked "Supervised ✗", while CORRECT is marked ✓.

- In the transfer experiment (C.5), the configuration is re-selected on target labels, pooling the target's reference and validation files (50% of the target data). Figure 4(c) therefore does not show label-free generalization.
- The selected bands and layers are never reported, nor whether they are stable across seeds and datasets.
- The needed experiment is a fixed, label-free configuration, or a configuration chosen on one dataset and frozen on the others.

2. The propagation component contributes little outside WW-AG, and the qualitative examples do not show what the text claims.

- Rescoring adds +8.5 and +6.9 points on WW-AG. Elsewhere it adds +1.15 and 0 on WW-HC, +2.3 on TE-Cap, +2.2 and +0.7 on TE-Mag, and +0.4 on CE. That is about one test trajectory per split or less.
- Table 4 covers only WW-AG and CE. On CE, all variants lie within 61.4–61.8.
- In the paper's featured example (Figures 6 and 12), every non-final step receives a near-identical boost of +0.08 to +0.09, and the final step receives zero by construction ($B_T = 0$). The base margin being overturned is 0.88 vs. 0.87. This is a last-step penalty, not signal routed "four steps" back.
- The WW-HC example works the same way: the displaced step S73 is the final step and cannot be boosted.
- In Figure 13, S3 and S6 tie at 1.44 at the printed precision, and the stated tie rule would pick the wrong one.
- With $w=1$, which is selected on WW-AG for both backbones, propagation mostly credits whichever step the next step attends to most. The missing control is "add $\gamma \cdot S(s_{i+1})$ to step i ", along with "exclude the final step".
- Raw attention mass summed over key tokens is biased toward long steps and toward step 1, which contains the task description and the attention sink. No length normalization or faithfulness check is given.
- The preferred attention band is the earliest on Qwen and the latest on DeepSeek, so the interpretation that "lower blocks give clearer dependencies" is post hoc.

3. The central premise, that failed trajectories are what matter, is untested and partly contradicted by the paper's own results.

- The synthetic reference in A.7 "mixes successes and failures" yet nearly matches the benchmark reference (46.56 vs. 47.62).
- Six to twelve reference trajectories suffice, and references from other datasets transfer.
- Together with the use of only the top 20 directions of an uncentered matrix, this suggests R captures generic representation geometry rather than failure structure.
- The comparison with OAT confounds the method with the data: OAT is trained on 103 out-of-distribution MCP-Atlas trajectories, while SOAP gets in-distribution references and labeled validation.
- The needed controls are an R fit on success-only trajectories and an R fit on unrelated text.
- The Huber mixture in Section 2 is not used by the method. No argument is given for why error steps should have low energy in a middle band.

4. Trivial baselines and obvious confounds are missing.

- The score is an unnormalized projection of a mean-pooled vector, so it depends on the vector norm, and hence plausibly on step length and agent type. In Figure 11, "Next speaker" turns score about 0.9–1.1, instructions about 1.5–1.9, and long WebSurfer page dumps score highest (2.0–2.65).
- On WW-AG, decisive steps sit early in trajectories capped at 10 turns (mean index 3.0).
- The paper should report fixed-index, first-non-orchestrator-step, step-length and agent-identity baselines.
- It should also report a linear probe or few-shot judge given the same validation labels SOAP uses.

5. Baseline fairness.

- StepFinder's encoder is replaced by the first 128 and 32 coordinates of a raw LLM hidden state. This is reasonable for a Matryoshka embedding model but arbitrary for a hidden state, and it likely cripples the baseline.
- Neither training-based baseline is tuned on the validation split.
- Bolding "within block" hides that GPT-4o ErrorProbe beats SOAP-Qwen on TE-Mag (28.26 vs. 23.19). The main-text claim that the advantage "persists with both GPT-4o and GPT-5" is contradicted by Tables 1 and 15.

6. Statistical support is thin.

- Test sets contain 29–63 trajectories. Three random splits of the same pool overlap heavily, so the seed standard deviation understates uncertainty.
- The standard-deviation convention also differs between Table 13 and Table 16 (± 2.59 vs. ± 3.2 for the same cell).
- Paired tests or bootstrap intervals over trajectories are needed.

7. Related work. The formulation is very close to HaloScope (Du et al., NeurIPS 2024), which uses a Huber contamination mixture over unlabeled LLM representations and SVD-subspace scoring. It is also close to spectral signatures (Tran et al., 2018). Neither is cited. The literature on attention faithfulness and attention sinks is also absent.

Minor issues.

- The scalability sentence compares SOAP on the 27B model (44.97) with StepFinder on the 14B model (22.65).
- B.2 says non-WW-AG cells peak at $\gamma \in \{0.1, 0.2\}$, but Figure 13 implies $\gamma = 1.0$.
- The notation alternates between q_C and π_C .
- Figure 3's axis is labeled "base score" but shows projection energy.
- Eq. 6 is 0/0 for steps that no later step retains as a predecessor.
- Table 7 reports seconds per forward pass, while Table 8 reports the same numbers as seconds per trajectory. Scoring requires per-step contexts under truncation and materialized attention maps.

Questions for the authors

- 1 What is the accuracy with one label-free or cross-dataset-frozen configuration?
- 2 Which bands and layers were selected, and how stable are they across seeds and datasets?

- 3 What are the results for the controls in weaknesses 2–4: next-step shift, final-step exclusion, a success-only or unrelated-text reference, and length, position and agent baselines?
- 4 What is the end-to-end wall-clock time per trajectory on WW-HC?

Rating

- Soundness: 2/4
- Presentation: 3/4
- Contribution: 2/4
- **Overall: 4, marginally below the acceptance threshold** (assuming the ICLR 2026 scale)
- Confidence: 4/5

The method is simple and the numbers are striking. However, the evidence does not yet support the three claims the paper rests on: that the method is free of step-level labels, that failure-conditioned references are the source of signal, and that attention propagation resolves downstream contamination. I would raise my score if the label-free protocol and the controls above hold up.